

Manifold-Aware Anomaly Detection & Generative Drifting

Weekly Log & Progress Report

24 June 2026

Key Objectives Completed:

- **Investigated K-NN Breakdown:** Analysed the mathematical limits and "instability" of K-Nearest Neighbours in high-dimensional spaces.
- **Established Baseline:** Finalised a K-NN anomaly detection baseline using PatchCore.
- **CDC Implementation:** Applied the Carré-du-Champ (CDC) operator to explicitly measure tangential vs. orthogonal flow.
- **Generative Drifting:** Designed an annealed tangential repulsion field to prevent generative mode collapse while respecting manifold topology.

The "Curse of Dimensionality" in K-NN

Distance Concentration Theorem (Beyer et al., 1999)

Condition: As dimensionality (m) approaches infinity, if the variance of the scaled distance distribution converges to zero:

$$\lim_{m \rightarrow \infty} \text{var} \left(\frac{(d_m(P_{m,1}, Q_m))^p}{E[(d_m(P_{m,1}, Q_m))^p]} \right) = 0$$

Result: Then for every $\epsilon > 0$, the maximum distance ($DMAX_m$) converges to the minimum distance ($DMIN_m$):

$$\lim_{m \rightarrow \infty} P \left[DMAX_m \leq (1 + \epsilon) DMIN_m \right] = 1$$

The Implication:

All data points become virtually equidistant to the query point, K-NN loses its structural meaning.

Why Distance Contrast Collapses (1/2)

1. Distance as a Sum of Components:

Assume an IID scenario. The total distance d_m across m dimensions is the sum of the distance contributions (X_i) from each individual dimension:

$$d_m = \sum_{i=1}^m X_i$$

2. The Growth of the Expected Value:

Since all dimensions are IID, let the expected distance in one dimension be μ . The expected total distance is:

$$E[d_m] = m\mu$$

$$(E[d_m])^2 = (m\mu)^2 = m^2\mu^2$$

Why Distance Contrast Collapses (2/2)

3. The Growth of the Variance:

The variance of a sum of independent variables is the sum of their variances. If one dimension has variance σ^2 , the total variance (the numerator) grows only **linearly**:

$$\text{var}(d_m) = \sum_{i=1}^m \text{var}(X_i) = m\sigma^2$$

4. The Convergence to Zero:

We evaluate the variance of the *scaled* distance as $m \rightarrow \infty$:

$$\lim_{m \rightarrow \infty} \frac{\text{var}(d_m)}{(E[d_m])^2} = \lim_{m \rightarrow \infty} \frac{m\sigma^2}{m^2\mu^2} = \lim_{m \rightarrow \infty} \frac{\sigma^2}{m\mu^2}$$

Because σ^2 and μ^2 are fixed constants, therefore m in the denominator drives the entire fraction to **0**.

PatchCore Baseline Results (MVTec AD)

Baseline Performance Thresholds:

Category	Image AUROC	Patch AUROC
bottle	1.0000	0.9856
cable	0.9906	0.9850
capsule	0.9912	0.9901
carpet	0.9868	0.9907
grid	0.9891	0.9835
hazelnut	1.0000	0.9884
leather	1.0000	0.9923
metal_nut	0.9971	0.9870
pill	0.9463	0.9813
screw	0.9596	0.9892
tile	1.0000	0.9554
toothbrush	0.9194	0.9889
transistor	0.9942	0.9727
wood	0.9860	0.9314
zipper	0.9756	0.9813

Performance Comparison: Baseline vs. Tangential Repulsion

Initial Results: Annealed Tangential Repulsion (ν_{ortho})

Category	PatchCore (Image AUROC)	New Method (ν_{ortho})
Bottle	1.0000	1.0000
Grid	0.9891	0.9983
Metal Nut	0.9971	1.0000
Tile	1.0000	1.0000
Toothbrush	0.9194	0.9917
Wood	0.9860	0.9982